

Transformer Encoder — Self-Attention Model

Per-model deep dive · the best single architecture and the hybrid's namesake

Result: 0.522 accuracy · 0.502 weighted F1 · WER 0.478 · fold SD ± 0.149

Notebook: 05_transformer_wlasl_landmark_baseline.ipynb

1 Theory

A Transformer encoder uses **self-attention**: every frame computes a weighted combination of every other frame, with the weights learned from the data. Unlike the LSTM (which passes information step-by-step) or the ST-GCN (which uses a fixed graph), attention lets any frame relate directly to any other, at any distance, with no recurrence and no hand-specified structure. It is the most flexible temporal model — and the architecture the thesis's 'Hybrid Multimodal Transformer' is built around.

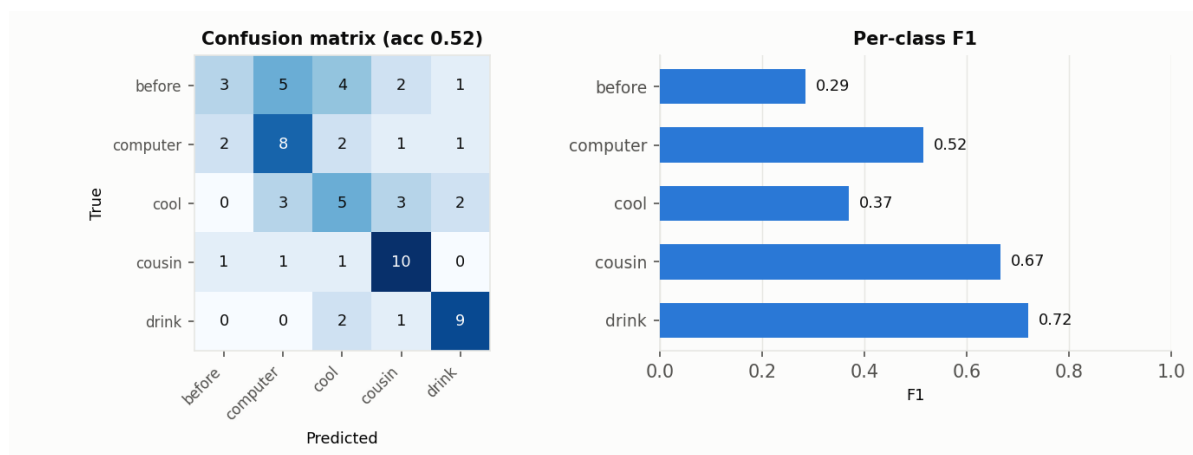
2 Method of execution

The 1,629-D frames are linearly projected to a 128-D model space and given **sinusoidal positional encodings** (so the otherwise order-agnostic attention knows frame order). A 2-layer, 4-head Transformer encoder (feed-forward 256, GELU, dropout 0.3) processes the sequence with padding masked out; the valid frames are mean-pooled, LayerNorm-ed, and classified by a linear head.

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frames(1629) → Linear(128) → +positional encoding → TransformerEncoder(2 layers, 4 heads)
→ masked mean-pool → LayerNorm → Linear → 5 classes
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3 Results

Metric	Value (5-fold CV)
Accuracy	0.522
Weighted precision	0.518
Weighted recall	0.522
Weighted F1	0.502
Word Error Rate (WER)	0.478
Fold accuracies	0.29, 0.43, 0.62, 0.69, 0.62
Cross-val SD	± 0.149
Distinct classes predicted	5 of 5
Majority baseline	0.224



Transformer confusion matrix and per-class F1 under 5-fold CV.

4 Why these results

The Transformer is the **best individual architecture** at 0.522. Attention captures the whole-sign temporal structure — the relationship between the wind-up and the stroke, however far apart — that averaging (RF) and stage-independence (HMM) cannot. Its fold spread is high (SD ± 0.149 ; folds 0.29, 0.43, 0.62, 0.69, 0.62), because attention is data-hungry and 54 videos under-determine it, but its central tendency is the strongest of the single models. As data grows it is the architecture with the most headroom.

What makes it stronger / weaker

Stronger: the most expressive temporal model, no fixed structural assumptions, the natural backbone for a full KSL system. **Weaker:** the highest appetite for data of any model here — its variance will only come down with more examples.

5 Role in the hybrid — a core component

The Transformer is **the second stream of the recommended hybrid**. It supplies temporal expressiveness — the ability to weigh any frame against any other — which it pairs with ST-GCN's anatomical stability. The two are complementary: attention is powerful but high-variance, the graph is stable but structurally constrained, and jointly training their fused features lets the model take the best of both. This is the empirical and architectural heart of the 'Hybrid Multimodal Transformer'.